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VISUAL FLOWCHART

Spring Boot Request Lifecycle

→ Swipe through all 6 slides



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Step 1: Tomcat Receives Request

Embedded server catches HTTP

HTTP Request arrives at port 8080

Embedded Tomcat:

- Accepts TCP connection
- Parses HTTP (method, headers, body)
- Creates `HttpServletRequest` object
- Hands off to Spring's `DispatcherServlet`

💡 Tomcat is embedded in Spring Boot — no separate install



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Step 2: Filter Chain

Security, CORS, logging — before your code

Request passes through filters:

1. CorsFilter → CORS headers
2. SecurityFilter → JWT validation
3. LoggingFilter → request logging
4. Custom filters you add

If auth fails → 401 returned here

Request never reaches your controller

💡 Filters run BEFORE your controller code



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Step 3: DispatcherServlet

The traffic cop of Spring MVC

DispatcherServlet:

- Receives request from filters
- Asks HandlerMapping:
'Which controller handles GET /users/1?'
- HandlerMapping returns:
UserController.getUser()
- Dispatches request to controller

💡 DispatcherServlet = central router for all requests



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Step 4: Controller → Service → Repo

Your code runs here

```
@GetMapping("/users/{id}")
public UserDTO getUser(@PathVariable Long id) {
    return userService.findById(id);
}
```

- Service layer: business logic
- Repository layer: database query
- Hibernate: generates SQL
- MySQL: returns data
- Entity → DTO → returned

💡 Controller → Service → Repository. Always.



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Step 5: Response Serialization

Java object → JSON

UserDTO object returned from controller

Jackson (default serializer):

- Converts Java object to JSON
- Respects @JsonIgnore, @JsonProperty
- Sets Content-Type: application/json

```
{  
  "id": 1,  
  "name": "Shubham",  
  "email": "shubham@test.com"  
}
```

💡 Jackson handles all serialization automatically



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Step 6: Response Sent

Back through the chain

Response travels BACK through:

- Exception handlers (if error)
- Response filters
- Tomcat sends HTTP response

HTTP/1.1 200 OK

Content-Type: application/json

{ ... }

Total time: ~5-50ms typically

💡 Understanding this lifecycle = understanding Spring Boot



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done = true

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